# Clarissa Morgan

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## Education

**California State University, Fresno** — Fresno, CA
*B.S. Computer Engineering | GPA: 3.7 | President's List (2 semesters)* — *Aug. 2024 – Expected Dec. 2027*

**Lemoore College & Lemoore Middle College High School** — Lemoore, CA
*Earned concurrently: A.A. Mathematics & Science, A.A. Arts & Humanities, H.S. Diploma* — *Aug. 2020 – May 2024*

## Experience

**Computer Engineering Intern** — May 2026 – Aug. 2026
*Intel Corporation* — *Hillsboro, OR*

- Built an AI analytics system (Python, SQL, Polars) that analyzes fab data through Intel's internal LLM gateway, generating risk analysis, predictions, and insights; cut an estimated 75% of manual analysis work
- Made it work with any database and split large datasets (1,000+ columns, 10,000+ rows) into chunks the model processes reliably, keeping all data inside Intel's network (OAuth2, SSL verification)
- Built a tool-inventory dashboard (Power BI, TOAD Data Point, Docker) saving its team ∼7 hrs/week, and a parallelized utility that resolves FQDNs and pings fab hosts in batches, saving ∼2 hrs/week
- Prototyped RAG and MCP integrations with internal AI tools; documented the RAG setup for other employees

**Automotive Ethernet Research Assistant** — Jan. 2026 – Present
*California State University, Fresno* — *Fresno, CA*

- Characterize voltage profiles of automotive Ethernet networks and develop cybersecurity measures for them

**AI & Robotics Research Assistant** — May 2025 – July 2025
*California State University, Fresno* — *Fresno, CA*

- Upgraded a 6-DoF strawberry-harvesting arm's vision (Python, NVIDIA Jetson AGX Orin) from 2D color detection to 3D detection that measures each strawberry's distance so inverse kinematics can guide the arm
- Discovered the arm's camera stream stayed accessible while powered off; documented the vulnerability

## Projects

**Hybrid Counting Sort with a Balanced BST** | *C++* — Feb. 2026 – May 2026

- Designed a sort that counts repeated values in a balanced BST: $O(n \log d)$ time, $O(d)$ space ($d$ = distinct values)

**Network Attack Lab: DoS & Packet Replay** | *hping3, nginx, VirtualBox, Linux* — Jan. 2026 – Present

- Ran and analyzed packet-replay and DoS port-flooding attacks on a Minecraft server in an isolated VM lab

**AES-128 Core with Hardware Trojan** | *Verilog, FPGA* — Fall 2026

- Co-designed a synthesizable AES-128 core in Verilog; verified encryption output against NIST test vectors
- Inserted a hardware trojan that corrupts output only on a hidden trigger; verified in simulation and on FPGA

**Nios V (RISC-V) Embedded System** | *C, Quartus Prime Pro, Platform Designer* — Course Project

- Built a Nios V system with peripherals; programmed board I/O in C using timers and interrupts

## Leadership & Teaching

**Chair (President)** — Jan. 2025 – Present
*IEEE Fresno State Student Branch* — *Fresno, CA*

- Grew events from ∼5 to 30+ per semester, including PCB, soldering, and Python workshops for 50+ students
- Re-established ties with our IEEE section and area, then secured $6,000 to send 11 members to IEEE Rising Stars
- Brought Fresno State into the IEEE Elevate pilot, funding senior capstone work at an under-resourced campus, and revived the Micromouse robot competition in partnership with the Robotics Club

**Founder & President** | *Lemoore Middle College High School Peer Tutoring Program* — Apr. 2021 – May 2024

- Founded a peer tutoring program and led a team of 4 tutors in delivering 500+ hours of academic support
- Raised tutored students' grades by 1–2 letter grades on average, helping at-risk students recover from failing

## Technical Skills

**Languages**: C, C++, Python, SQL (PostgreSQL), Verilog, ARM Assembly, MATLAB, PowerShell
**Hardware**: Nios V (RISC-V), FPGAs (DE1-SoC, DE25), Jetson, Raspberry Pi, Vivado, ModelSim, oscilloscopes
**Security**: Threat modeling (STRIDE, DREAD), OWASP IoT, Linux hardening
**AI & Data**: LLM integration, RAG, MCP, Polars, pandas, Power BI (DAX, Power Query), TOAD Data Point, cnvrg.io
**Systems & Networking**: Linux, Git, Docker, VirtualBox, Tailscale, SSH, nginx, nmap, hping3, OAuth2